# Loan Default Intelligence

Simple guide to the Customer Behavioral Analysis dashboard

This guide explains what each dashboard board shows, how to read the graphs, and how to use the loan review simulator.

**Data:** PKDD'99 historical banking data
**Portfolio:** 682 loan applications, including 76 defaults
**Default rate:** 11.1%

**Important:** The dashboard is a decision-support and learning tool. It is not a replacement for affordability checks, identity checks, current credit data, human review, or fair-lending controls.

# 1. What the dashboard does

The dashboard takes historical customer banking behaviour and shows which patterns are associated with loan default. It combines data preparation, charts, machine-learning model comparison, a scenario simulator, and practical recommendations in one place.

| Board | Purpose in simple words |
|---|---|
| Loan review simulator | Try a loan scenario and see the risk level, model class, and recommended next action. |
| Model comparison | Compare six machine-learning models and see which one ranks risk best. |
| Project overview | Understand the whole project from raw banking data to recommendations. |
| Data & pipeline | See how raw records become clean, pre-loan loan features. |
| EDA & portfolio signals | Explore defaults, balances, behaviour, and loan duration. |
| FP-Growth & patterns | Look for combinations of customer behaviours that occur together. |
| Validation test | See how the model was tested and whether results are reliable. |
| Model innovation | Compare the headline model with a capacity-aware experiment. |
| Recommendations | Translate findings into responsible credit-team actions. |

## How to use it

? Start with Project overview if you are new to the project.

? Use EDA & portfolio signals to understand the data before looking at model scores.

? Use Model comparison and Validation test to judge model quality.

? Use Loan review simulator for what-if analysis, not automatic approvals or rejections.

? Use Recommendations for the practical next steps and governance boundaries.

# 2. Loan review simulator

This is the most operational board. Enter a requested loan amount, loan duration, available balance, minimum observed balance, and transactions per month. Then choose the Class 1 / manual-review cutoff.

## How the result is decided

The simulator calculates a risk index from the entered values. It compares that risk index with the cutoff you select:

| If... | The simulator shows... |
|---|---|
| Risk score is below the cutoff | Class 0 ? Good, Lower risk, and normal approval checks can continue. |
| Risk score is equal to or above the cutoff | Class 1 ? Default risk, Higher risk, and manual credit-officer review. |

**Example:** If the risk score is 11% and the cutoff is 45%, 11% is below 45%, so the result is Class 0 ? Good. If you lower the cutoff to 10%, the same 11% score is at or above the cutoff, so it becomes Class 1 ? Default risk.

## What the cards mean

? Risk level: an easy-to-read label, Lower risk or Higher risk.

? Recommended action: normal approval checks or send to a credit officer for manual review.

? Model class: the technical label, Class 0 or Class 1.

? Risk score: the calculated percentage used in the cutoff comparison.

? Exposure if defaulted: the requested amount, shown as the possible exposure.

? Expected loss proxy: requested amount multiplied by the risk score. It is an estimate, not a guaranteed loss.

The ?Why this result?? message highlights simple reasons, such as a below-median balance or an above-median requested amount. The simulator is intentionally transparent and should be used to request documents or affordability evidence, not to auto-reject someone.

# 3. Model comparison

This board compares Decision Tree, Naive Bayes, Random Forest, SVM, KNN, and Gradient Boosting. It lets you choose the metric, whether SMOTE balancing was used, and whether the model was tuned.

## How to read the graphs

? Metric heatmap: darker or stronger cells indicate better scores for the selected metric.

? Leaderboard: orders the models from best to worst for your selected metric.

? F1 versus ranking quality: F1 balances precision and recall; Average Precision and ROC-AUC show how well the model ranks higher-risk cases.

? All model runs: shows the individual experiment results rather than only the winner.

? Behaviour segments: connects model evaluation with the two customer behaviour groups.

? Portfolio risk lens: shows the practical trade-off between risk detection and portfolio impact.

The repeated cross-validation headline is tuned Gradient Boosting with F1 0.571 ? 0.120, Average Precision 0.654, and ROC-AUC 0.864. Tuned Random Forest is a close challenger and leads on Average Precision and ROC-AUC in the selection table. This means no model is perfect; the dashboard supports a shortlist rather than claiming certainty.

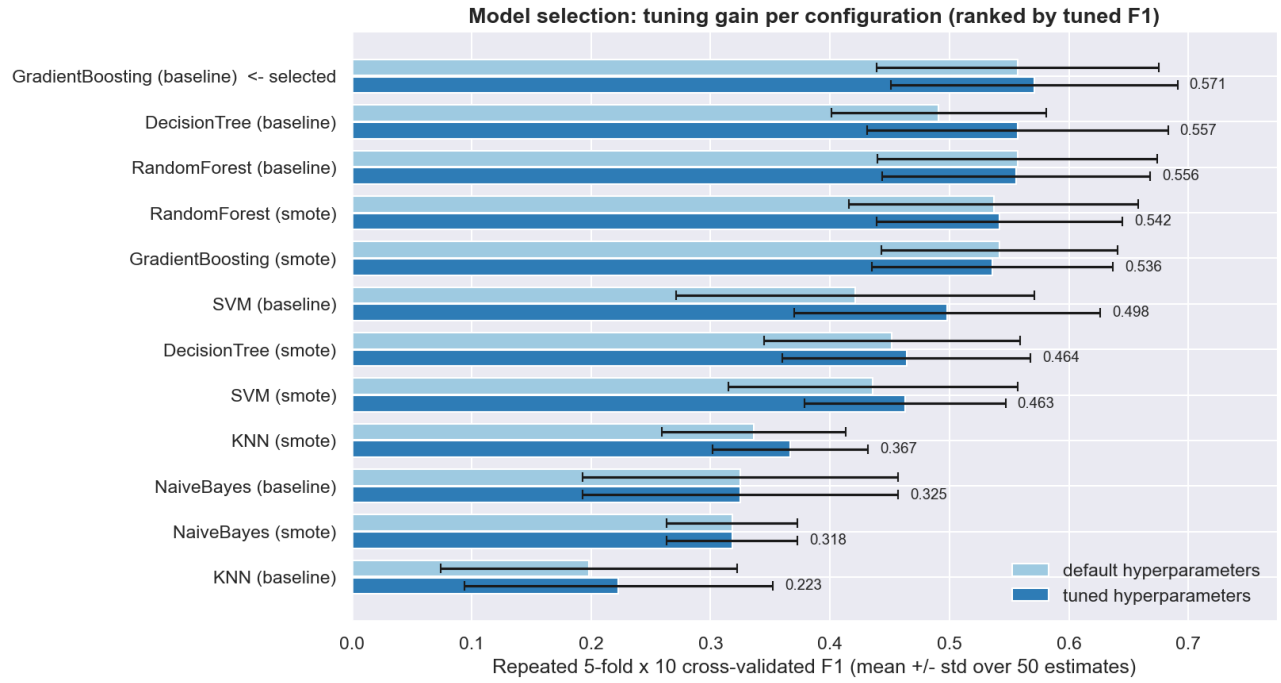



**Figure:** Repeated cross-validation model selection.

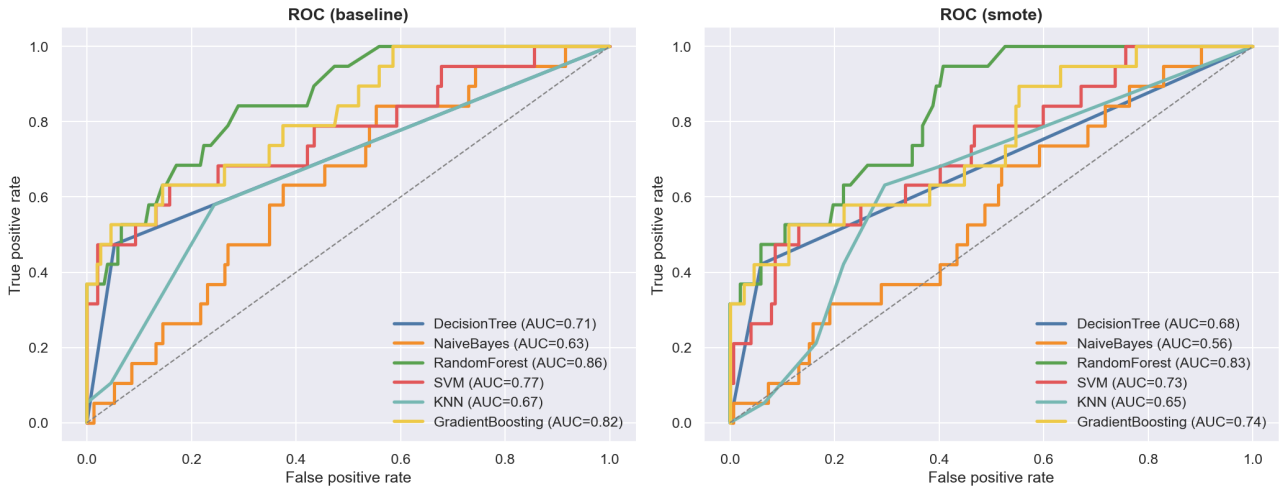



**Figure:** ROC curves: higher curves generally mean better ranking across thresholds.

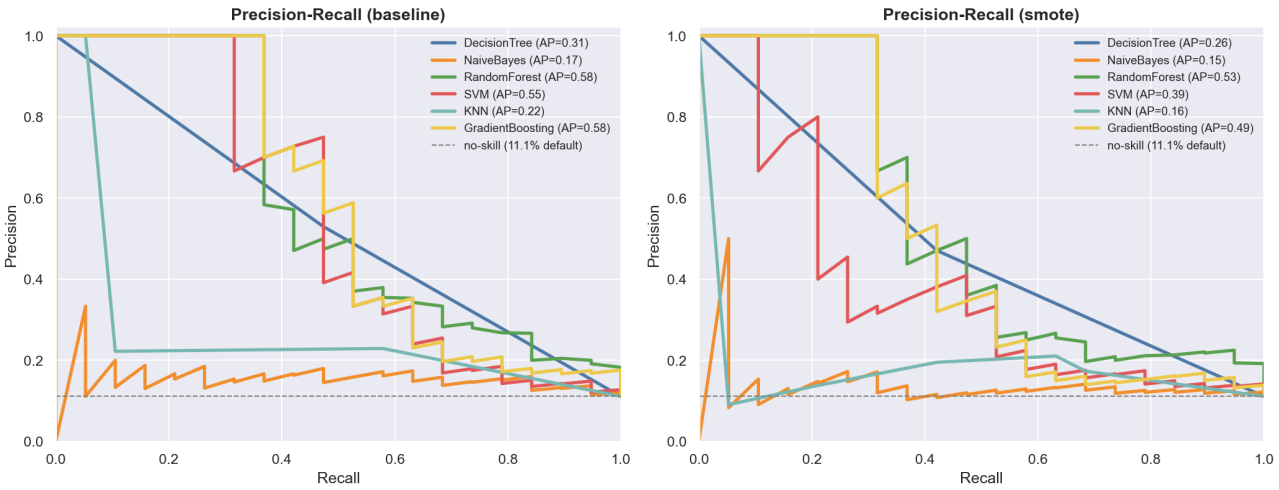



**Figure:** Precision-recall curves: useful when defaults are the smaller class.

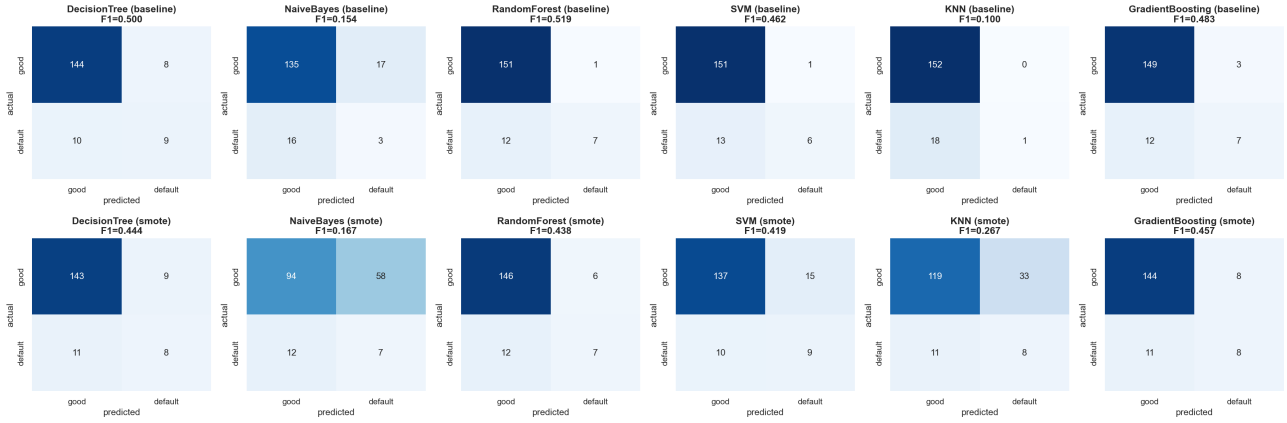



**Figure:** Confusion matrices: counts of correct and incorrect Good/Default predictions.

# 4. Project overview

This board explains the end-to-end story: raw loan, transaction, account, client, and disposition records are cleaned; pre-loan behavioural features are created; models are tested; segments and patterns are explored; and the results are turned into recommendations.

The project uses only information available before the loan date for its main features. That matters because using information that happened after a loan decision would make a model look better than it really is.

# 5. Data & pipeline

This board explains the source tables, preprocessing, feature engineering, and data-quality checks. The final feature table contains 682 loans and 27 columns, including loan terms, balances, transaction activity, and the target class.

? Class 0 means Good in this project.

? Class 1 means Default.

? The portfolio contains 606 Good loans and 76 Default loans.

? The target is imbalanced, so accuracy alone is not enough.

? Pre-loan features reduce the risk of target leakage.

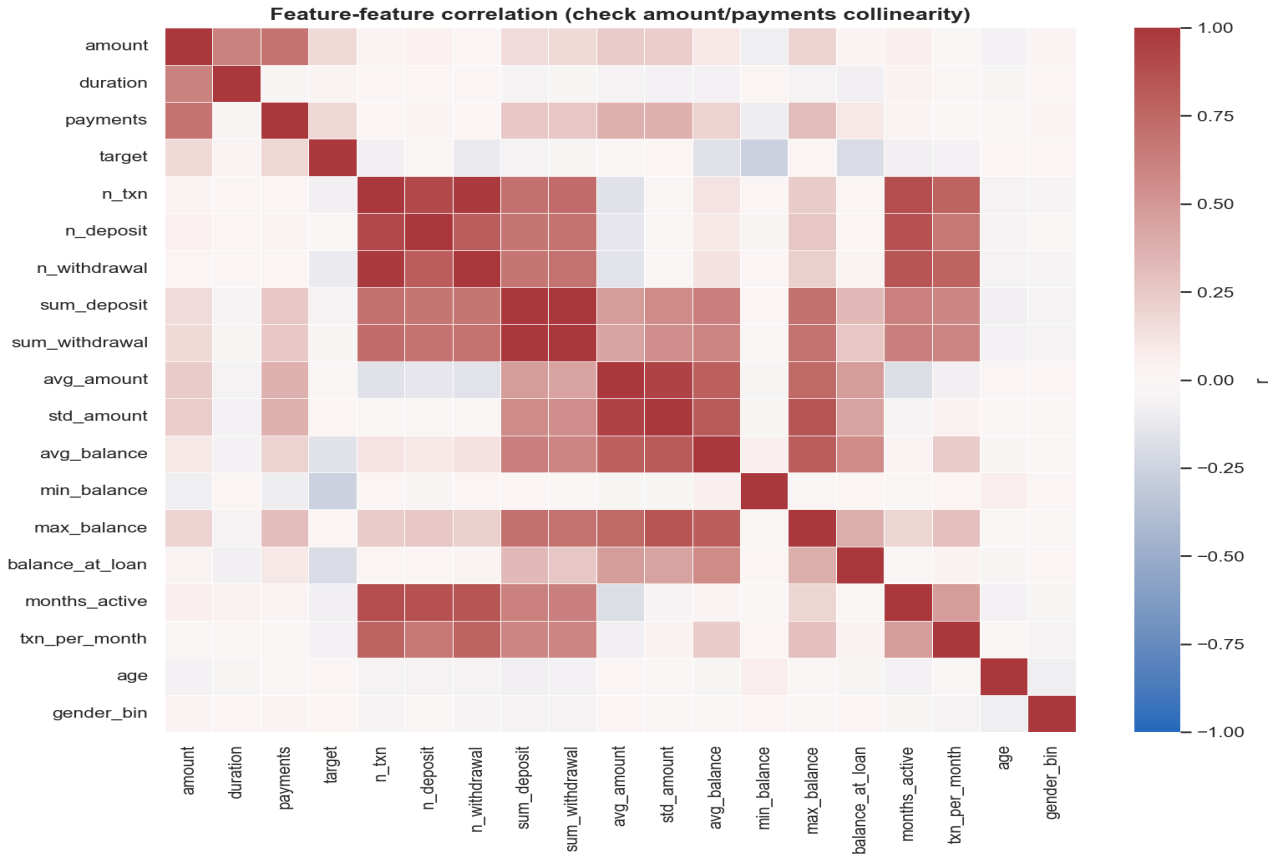



**Figure:** Correlation view across the engineered features.

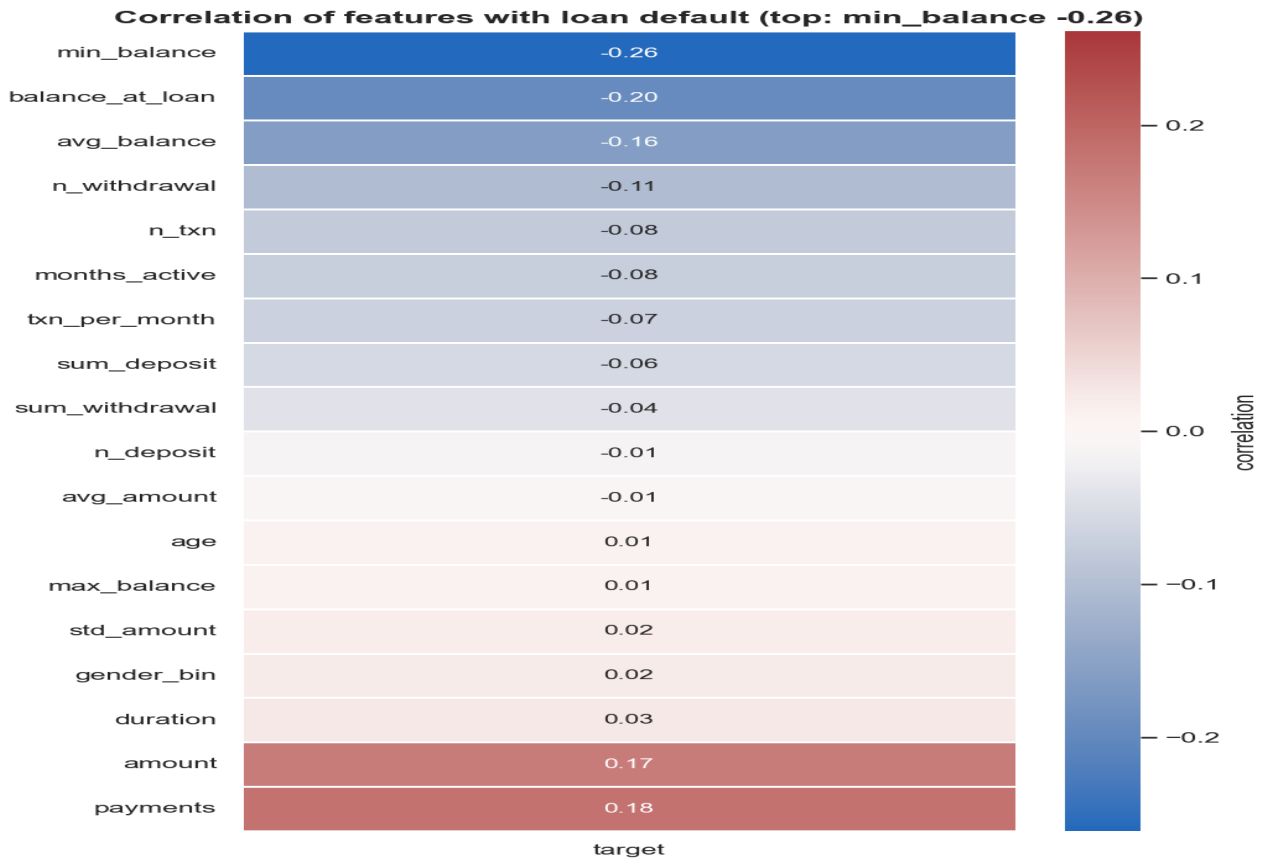



**Figure:** Relationship between features and the default target.

# 6. EDA & portfolio signals

This board is about understanding the portfolio before trusting a model. It shows how many loans are Good or Default, how balances differ between outcomes, and how default rates change by loan duration.

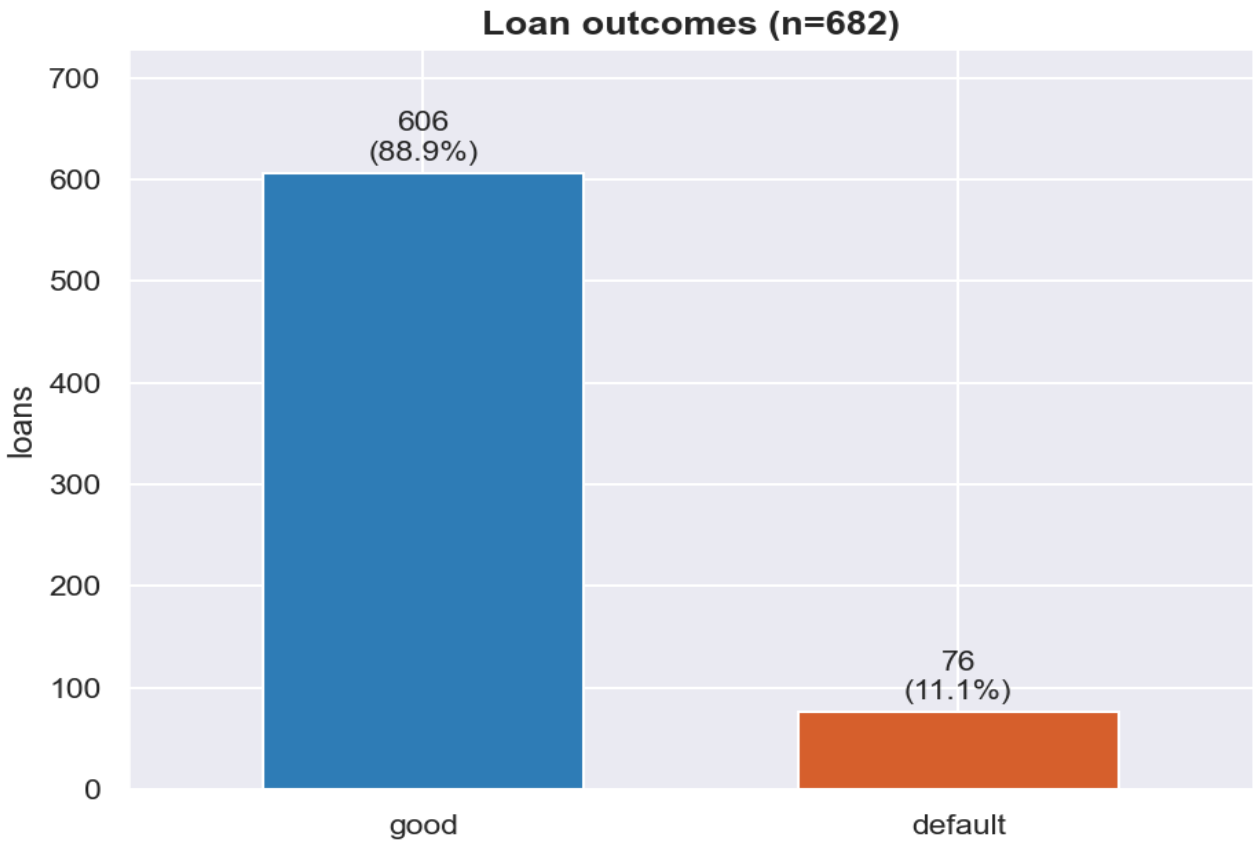



**Figure:** Portfolio composition: Good loans are the majority class and Defaults are the minority class.

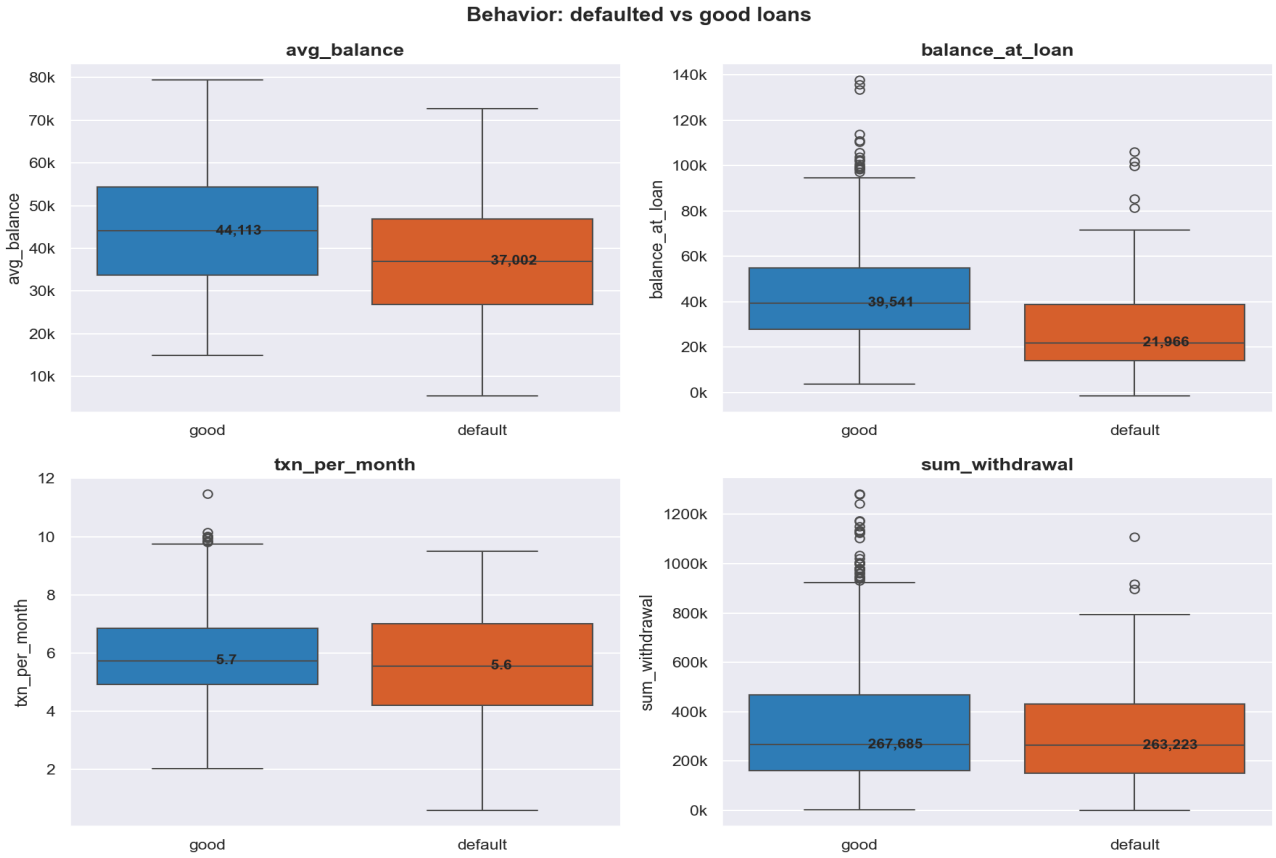



**Figure:** Behaviour distributions by outcome. Differences help identify useful signals, but overlap means no single feature decides every case.

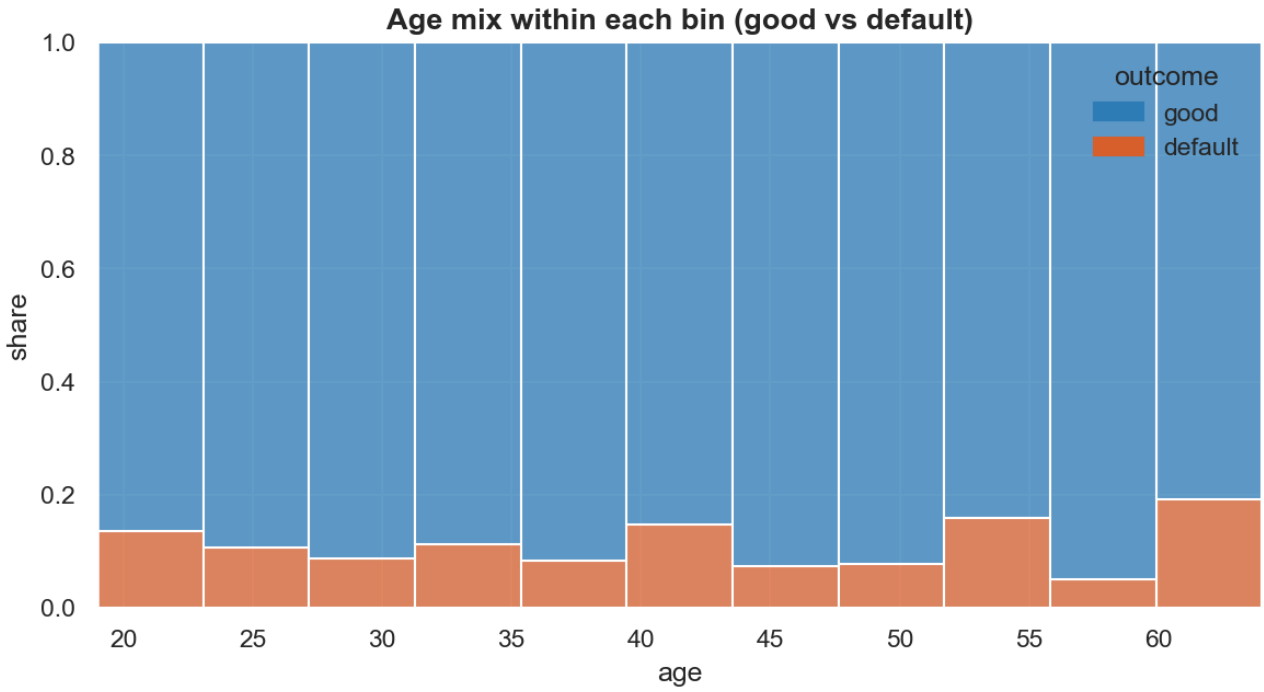



**Figure:** Age distribution of the modelling population. Age should not be used as an automatic rejection reason.

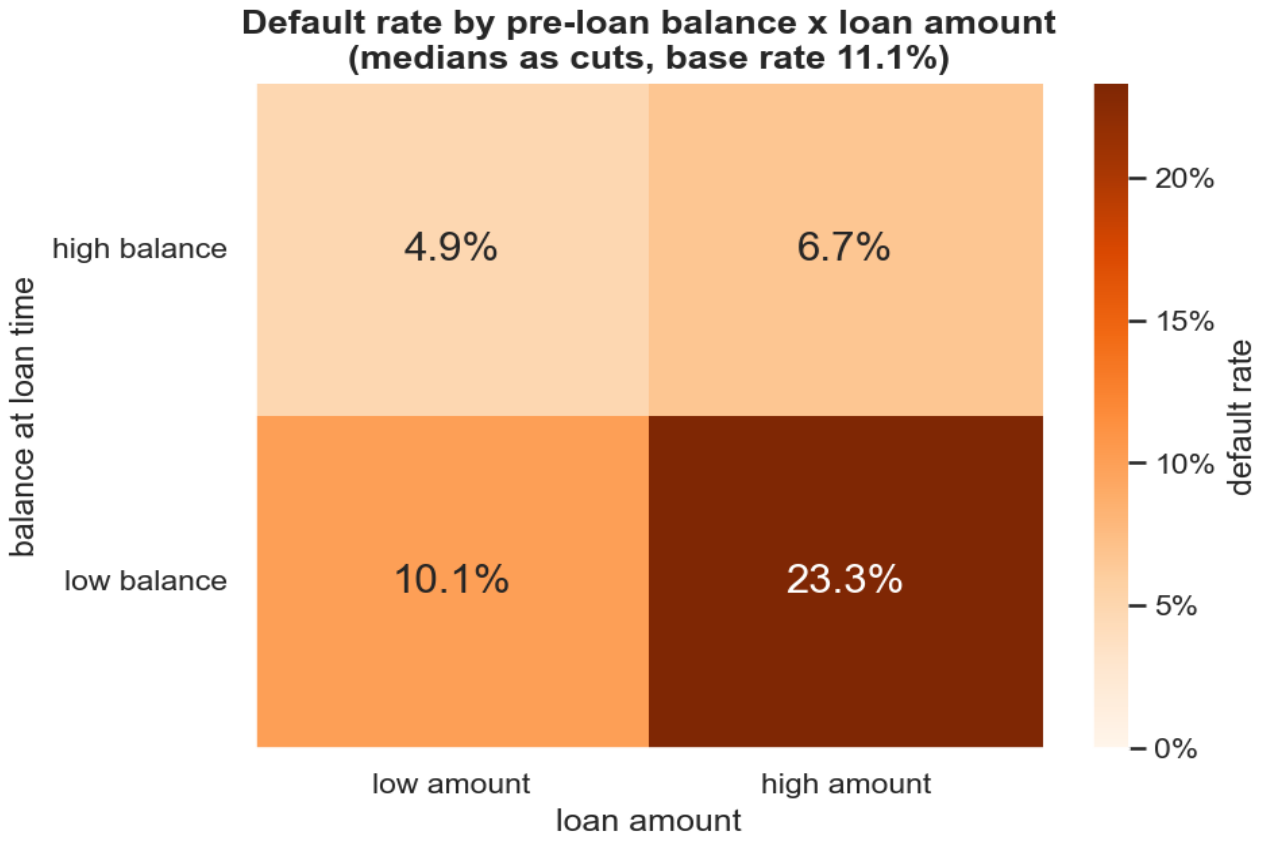



**Figure:** Risk quadrant: loan amount compared with pre-loan balance to show capacity pressure.

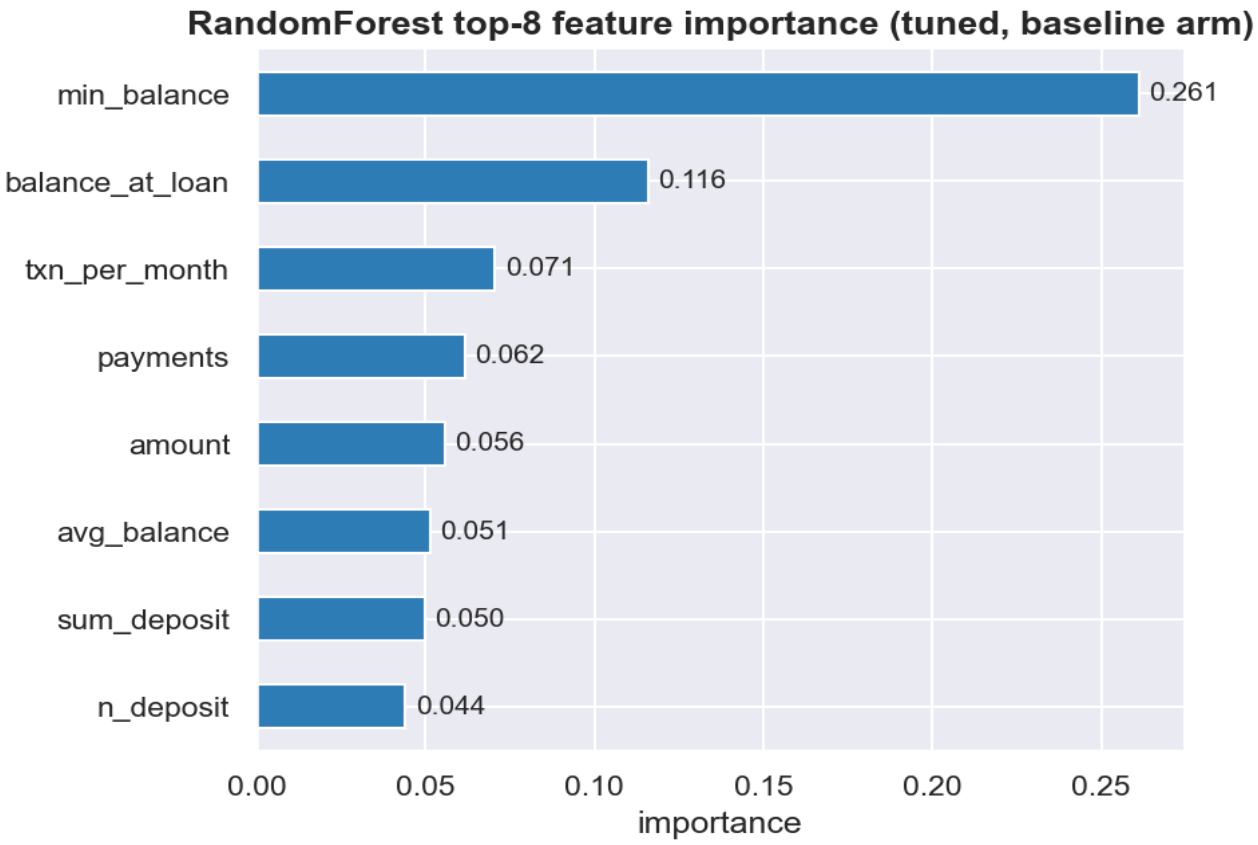



**Figure:** Random Forest feature importance. Balance-related features are among the strongest signals.

# 7. FP-Growth & patterns

FP-Growth looks for behaviours that commonly occur together, such as transaction or balance categories. It is descriptive: a frequent combination is not automatically a cause of default.

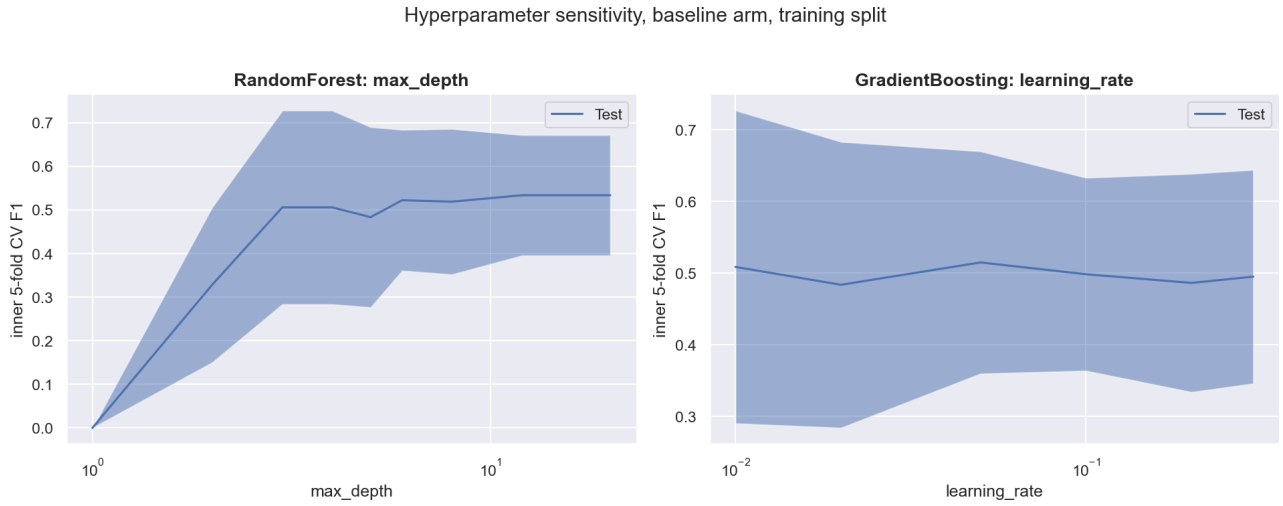



**Figure:** Tuning curves: how model performance changes during parameter search.

The board shows mining coverage, frequent patterns, and rules whose consequence is Default. In this project, no default rule met the configured confidence threshold, so pattern mining should support investigation rather than replace the classifier.

# 8. Validation test

Validation asks whether the model result is stable and meaningful. The project uses a held-out test set plus repeated stratified 5-fold cross-validation repeated 10 times. Stratification keeps the Good/Default mix more consistent across folds.

| Measure | Plain meaning |
|---|---|
| Accuracy | Overall percentage correct; can be misleading with an imbalanced target. |
| Precision | When the model flags Default, how often it is correct. |
| Recall | How many actual Defaults the model finds. |
| F1 | A balance between precision and recall. |
| Average Precision | Quality of ranking the positive/default class across thresholds. |
| ROC-AUC | How well the model separates the two classes across thresholds. |

The validation board separates a single held-out result from repeated cross-validation. A single split can look unusually good or bad; repeated cross-validation gives a better sense of variation.

# 9. Model innovation

The innovation board compares the headline model with a capacity-aware experiment. The extra idea is to consider loan-to-balance and payment-to-balance pressure, then test whether the decision threshold improves the practical result.

This is an experiment, not an automatic replacement for the selected model. A useful improvement must be checked for stability, calibration, fairness, and performance on current data.

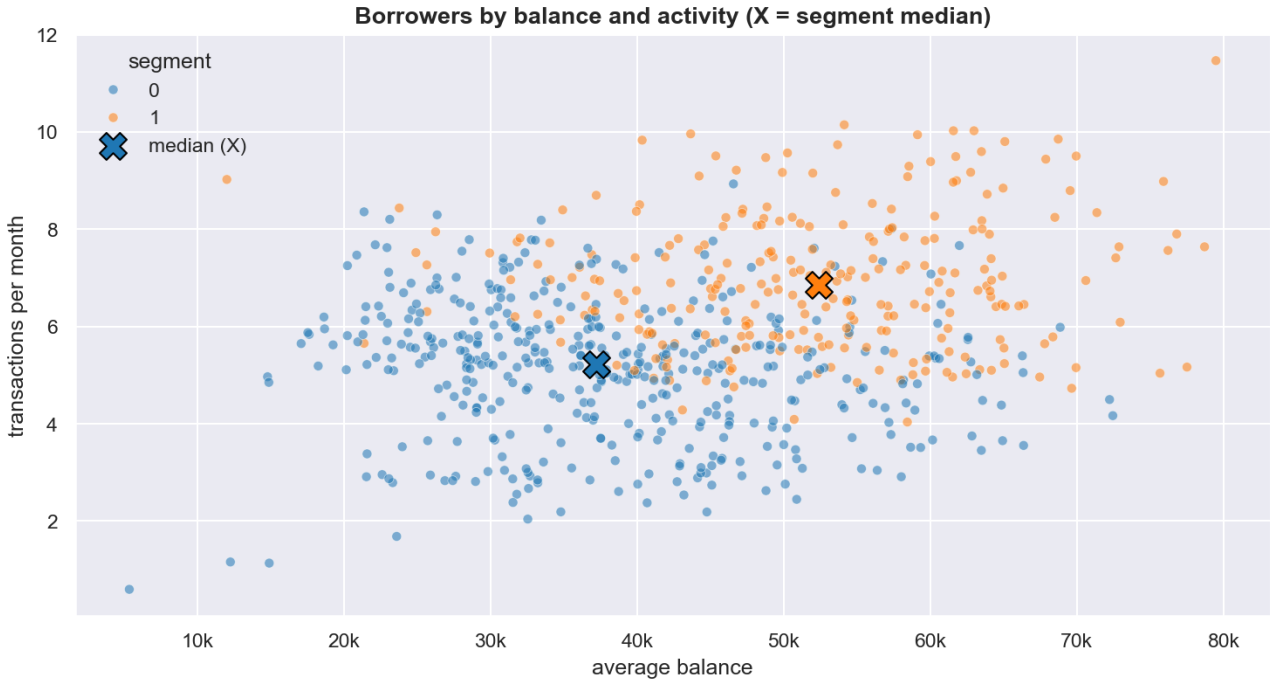



**Figure:** Behaviour segments shown as groups of customers based on transaction and balance behaviour.

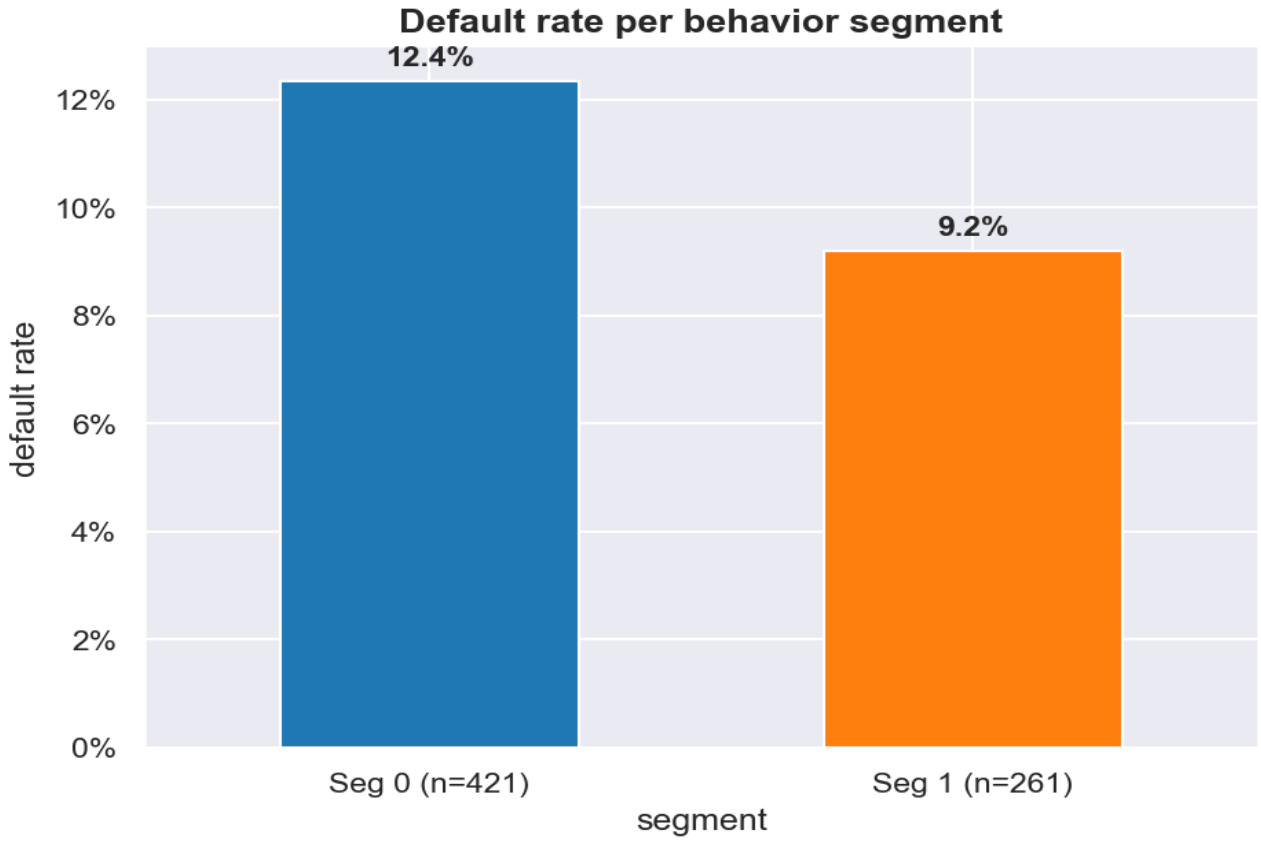



**Figure:** Default rate by behaviour segment.

The segmentation result has two groups: a quieter, lower-balance segment with about 12.4% default rate and a more active segment with about 9.2% default rate. The difference is useful for portfolio understanding, but it is not a standalone approval rule.

# 10. Recommendations

This board converts analysis into action:

? Credit review: prioritise low-balance and high-loan-to-balance cases for manual affordability checks.

? Portfolio policy: test different cutoffs in the simulator before changing approval policy.

? Model governance: revalidate on recent data and monitor recall, precision, calibration, and drift.

? Fair lending: do not use age or gender as automatic rejection reasons; use explainable financial-capacity signals.

? Future data: add affordability, bureau, repayment, and macroeconomic information when legally and operationally available.

# 11. Final interpretation

The dashboard does not answer ?approve? or ?reject? with certainty. It answers: ?Given these inputs and this selected threshold, should the case receive normal checks or additional human review?? That distinction is important because the historical data is limited, defaults are the minority class, and real lending decisions require current evidence and regulated controls.

**One-sentence takeaway:** Use the dashboard to understand risk, compare evidence, and prioritise review?not to make an automatic lending decision.